



Robot Language Acquisition

Based on Sequence-to-Sequence Learning

A Multimodal Machine Learning Approach to Human–Robot Communication

Dr. Ye Kyaw Thu

Lab Leader, Language Understanding Lab, Myanmar

Visiting Professor, Core-AI Lab, NECTEC, Thailand

Co-authors: K. Takabuchi, K. Fukai, N. Iwahashi, T. Kunishima

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Introduction & Motivation

Why Robot Language Acquisition?

Goal: build robots that **learn a natural language** from multimodal experience.

- Pre-programmed grammars are brittle and domain-locked
- Infants learn from **vision + speech + interaction** — robots should too
- Need grounded mapping between **concepts** (from video) and **spoken syllables**

This paper's claim:

Acquisition is possible **without any prior linguistic knowledge**, using sequence-to-sequence learning.

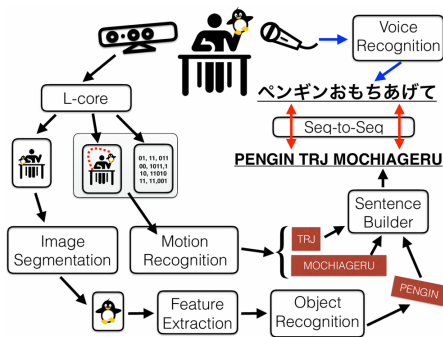


Fig. 1 — System overview

Contributions of This Work

- End-to-end pipeline: **motion video + user utterance** → language acquisition
- No prior linguistic or language-specific information
- Bidirectional translation between **Conceptual Structure (CS)** and **Syllable Sequence (SS)** of Japanese
- Four MT methods compared:
 - PBSMT, HPBSMT (statistical baselines)
 - Encoder–Decoder, Encoder–Decoder + Attention (neural)
- Trained / tested on **500 motion videos** with six motions and ten objects
- Neural models **match manually-prepared baselines** on several settings

Related Work

Related Work

- **Nakamura et al. [26]** — stochastic model of language & concepts from visual, haptic, audio, and word information
- **Spoken-word acquisition** — unsupervised clustering of speech tokens [11, 15, 34, 27]
- **Steels [37]** — socially-interactive evolution of grounded linguistic knowledge (game theory)
- **LCore / Iwahashi [16, 17]** — online learning of situated language from raw speech, vision, and tactile reinforcement

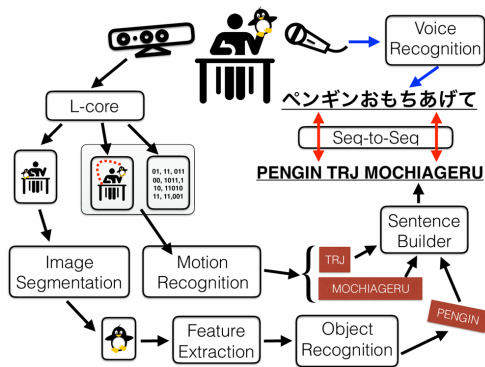
Our position: multimodal like [26] but *without haptic or word information*; uses sequence-to-sequence learning for CS $\leftrightarrow$ SS conversion.

Robot Language Acquisition System

System Overview

Pipeline components:

1. **L-Core** — image segmentation, motion recognition
2. **CNN feature extraction** (CAFFE + IMAGENET)
3. **Object recognition** — Gaussian-kernel SVM
4. **Motion recognition** — RDP-HMM
5. **CS generation** — heuristic
6. **Speech recognition** — Julius ASR $\rightarrow$ SS
7. **MT model** for $CS \leftrightarrow SS$



Example: CS = "PENGIN TRJ MOCHIAGERU"

SS = "pe-n-gi-n o mo-chi-a-ge-te"

Robot Language Acquisition System

L-Core

L-Core: In-house Language Learning Platform

- Designed for language learning from **multimodal inputs**: voice, images, motions [17, 28]
- Architecture: **client-server** with a central control program ptmove
- Connected servers: vision, speech, sound-quality-conversion, phonetic-typewriter, robot, Q&A
- Recently extended to drive the **Baxter Research Robot** via ROS
- In this work: used mainly for **image segmentation** and **motion recognition** on Kinect v1 videos

Robot Language Acquisition System

Object Recognition

Object Recognition Pipeline

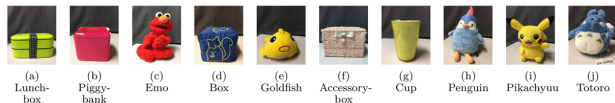


Fig. 8 — Ten objects used in the experiments

1. Image segmentation by L-Core
2. Feature extraction: [CAFFE](#) with open pre-trained [IMAGENET](#) model
3. Train classifier on extracted features
4. Best classifier: [Gaussian-kernel SVM](#) with ECOC multiclass decoding

Result: [98.7%](#) accuracy on 10 object classes (10-fold cross-validation, One-vs-One multiclass)

Math Refresher: Support Vector Machine

Binary SVM classifies a new point x by:

$$f(x) = \text{sgn} \left(\sum_i y_i \alpha_i K(x_i, x) + b \right)$$

- x_i — training samples; those with $\alpha_i \geq 0$ are the **support vectors**
- $y_i \in \{-1, +1\}$ — class label of x_i
- $K(x_i, x)$ — **kernel function** measuring similarity
- b — bias term

SVM is inherently a **binary classifier**. But our experiment requires classifying **10 object classes**.

Math Refresher: Gaussian (RBF) Kernel

To measure similarity $K(x_i, x)$, we use the Gaussian (RBF) kernel:

$$K(x, x_i) = e^{\gamma \|x - x_i\|^2}, \quad \gamma > 0$$

- $\|x - x_i\|^2$ — squared Euclidean distance between two points
- $K = 1$ when $x = x_i$ (identical points); $K \rightarrow 0$ as the points drift apart
- Smooth, infinite-dimensional feature space — well-suited to image features

To handle 10 classes, we use [Error-Correcting Output Codes \(ECOC\)](#) on top of binary SVMs (explained next).

From Binary to Multiclass: ECOC

SVM is inherently binary, but we have 10 object classes.

Error-Correcting Output Codes (ECOC):

- Assign each class a unique binary codeword (length $> \log_2 K$)
- Train one binary SVM per bit position
- At test time: predict each bit, then pick the class whose codeword is closest (Hamming distance)
- Error-correcting property:** individual SVM mistakes can be tolerated

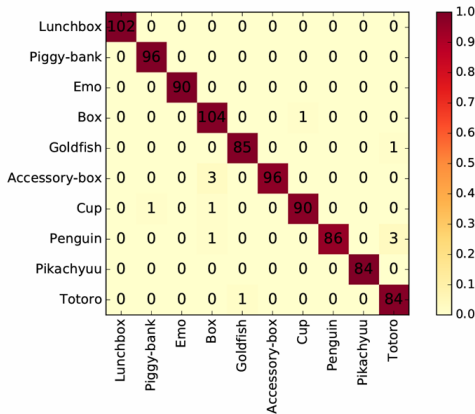


Fig. 2 — Confusion matrix,
Gaussian-kernel SVM, training data

Robot Language Acquisition System

Motion Recognition

Motion Recognition: Concepts

Words for objects are **nominal**; words for motions are **relational**.

A motion is a time-varying spatial relation between:

- **Trajector (TRJ)** — the moving entity (“figure”)
- **Landmark (LND)** — reference entity (“ground”)

Challenge: which object is the landmark, and which coordinate system is used, are **not observed** — the learner must infer them.

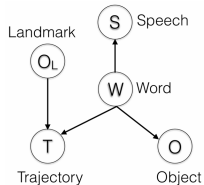


Fig. 3 — Graphical model of a lexicon over objects and motions

Math: RDP-HMM Motion Recognition

Notation:

- $V = \{v_i \mid i = 1, \dots, |V|\}$ — learned motion labels
- λ_i — HMM parameters for motion v_i
- k_i — index of the intrinsic coordinate system of v_i
- Y — observed trajectory of the moving object
- $\mathcal{R}$ — set of static object positions

Find the maximum-likelihood motion label $\hat{i}$ and reference point $\hat{r}$:

$$(\hat{i}, \hat{r}) = \arg \max_{i, r} P(Y \mid r, v_i, \mathcal{R}) = \arg \max_{i, r} P(Y \mid r, k_i, \lambda_i, \mathcal{R})$$

Result: 96.8% accuracy on 500 motions.

Only 16 errors — 2 are TRJ/LND confusion; 14 are HANASU (move-away) vs. CHIKAZUKERU (move-close).

Robot Language Acquisition System

Conceptual Structure & Syllable Sequences

Conceptual Structure (CS) Generation

Combine object IDs, motion IDs, and TRJ/LND tags *heuristically*.

Seven grammar patterns in this experiment:

1. OBJECT_LND OBJECT_TRJ HANASU
2. OBJECT_TRJ MAWASU
3. OBJECT_TRJ MOCHIAGERU
4. OBJECT_TRJ OBJECT_LND CHIKAZUKERU
5. OBJECT_TRJ OBJECT_LND HANASU
6. OBJECT_TRJ OBJECT_LND NOSERU
7. OBJECT_TRJ OBJECT_LND TOBIKOESASERU

Verbs: HANASU (move-away), MAWASU (move-circle), MOCHIAGERU (move-up), CHIKAZUKERU (move-close-to), NOSERU (move-onto), TOBIKOESASERU (move-over). Patterns 2 & 3 have no LND.

Syllable Sequence (SS) Recognition — Julius ASR

- Engine: [Julius](#) Japanese ASR [24]
- 500 sentences read by one native speaker
- Recorded in a noiseless environment

ASR accuracy

Syllable accuracy	88.6%
Sentence accuracy	15.4%

Output syllable sequences are used [directly](#) — errors included — in the seq-to-seq experiments.

Example ASR Errors (Table 1)

Reference	Hypothesis (Julius output)
chokinbako o koppu ni chikadukete erumo o komonoire ni nosete pikachuu o chokinbako kara hanashite bentoo o pikachuu no ue o tobikoesasete komonoire o mawashite	chokinbako o kopu ni chikazukete eruno komonoire ni nosete kikachuu o chokinbako kara hanashite bentoo o pikachu no uyo o tobikoesasete komonoire o mawasute

Substitutions, insertions, and deletions are all present. These noisy strings are fed directly to the MT models.

Methodologies

Four CS $\leftrightarrow$ SS Conversion Methods

1. **PBSMT** — Phrase-Based Statistical Machine Translation
 2. **HPBSMT** — Hierarchical Phrase-Based SMT (synchronous CFG)
 3. **Encoder–Decoder** — LSTM-based sequence-to-sequence
 4. **Encoder–Decoder + Attention** — soft attention over all encoder states
- **Source language:** Conceptual Structure (CS)
 - **Target language:** Syllable Sequence (SS) of Japanese
 - Both directions CS $\rightarrow$ SS and SS $\rightarrow$ CS are evaluated

Methodologies

PBSMT & HPBSMT

Phrase-Based SMT (PBSMT)

- Translation unit: **phrase** = contiguous sequence of words (not a linguistic phrase)
- Phrase table stores phrase-pair probabilities
- Lexicalized reordering model
- Decoder searches for the highest-scoring target sentence

```
BENTOO LND III べん と お III 0.310345 0.0274354 ...  
BENTOO LND III べん と お に III 1 0.252406 0.566667 ...  
BENTOO LND III ぺん と お III 0.13171 0.0102134 ...  
BENTOO LND III ぺん と お に III 0.852853 0.0939635 ...  
BENTOO TRJ III べん と お III 0.482759 0.349605 ...  
BENTOO TRJ III べん と お お III 1 0.349605 0.5625 ...
```

Fig. 4 — Sample phrase entries (CS → SS)

Hierarchical Phrase-Based SMT (HPBSMT)

- Based on **synchronous context-free grammar** (SCFG)
- Rules learned from un-annotated parallel text
- **Hierarchical structure** represents reordering explicitly
- Particularly strong for **long-distance reordering** [1]

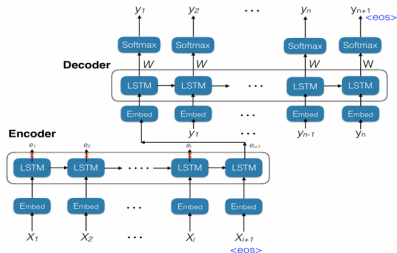
BENTOO LND [X] III べん と お [X] III 0.310345 0.0274354 . . .
BENTOO LND [X] III べん と お に [X] III 1 0.252406 0.566667 . . .
BENTOO LND [X] III べん と お [X] III 0.126784 0.0102134 . . .
BENTOO LND [X] III べん と お に [X] III 0.850649 0.0939635 . . .
BENTOO TRJ [X] III べん と お [X] III 0.482759 0.349605 . . .
BENTOO TRJ [X] III べん と お お [X] III 1 0.349605 0.5625 . . .

Fig. 5 — Sample HPBSMT grammar rules (CS $\leftrightarrow$ SS)

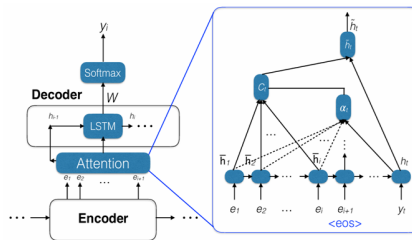
Methodologies

Sequence-to-Sequence Learning

Encoder-Decoder Model



(a) Encoder-Decoder translation model



(b) Encoder-Decoder with attention model

Fig. 6 — (a) Encoder-Decoder model, (b) Encoder-Decoder with Attention model

Encoder–Decoder: How It Works

Idea:

- Source RNN reads $x_1, x_2, \dots, x_m, \langle \text{eos} \rangle$
- Final hidden state = **fixed-length encoded vector** (“thought vector”)
- Target RNN generates $y_1, y_2, \dots, y_n$ from that vector
- Built with **LSTM** blocks

At each decoder step:

- Input: previous output y_{t-1}
- Update hidden state h_t
- Predict y_t via softmax

Weakness: compressing the entire source into a single vector is hard for long sentences.

Math: Encoder–Decoder Forward Pass

Encoder. Read source $\mathbf{x} = (x_1, \dots, x_m)$:

$$\bar{h}_t = \text{LSTM}_{\text{enc}}(x_t, \bar{h}_{t-1})$$

Final state $\bar{h}_m$ is the [thought vector](#).

Decoder. Generate target $\mathbf{y} = (y_1, \dots, y_n)$:

$$h_t = \text{LSTM}_{\text{dec}}(y_{t-1}, h_{t-1}), \quad P(y_t \mid y_{<t}, \mathbf{x}) = \text{softmax}(W_o h_t)$$

Train by maximizing the log-likelihood:

$$\mathcal{L} = \sum_{t=1}^n \log P(y_t \mid y_{<t}, \mathbf{x})$$

Encoder–Decoder with Attention

Key idea: the decoder can look back at all encoder states $\bar{h}_1, \dots, \bar{h}_m$, weighting them differently at each output step.

Step 1 — alignment weights (softmax over similarities):

$$\alpha_t(i) = \frac{\exp(\langle \bar{h}_i, h_t \rangle)}{\sum_{j=1}^m \exp(\langle \bar{h}_j, h_t \rangle)}$$

$\langle \cdot, \cdot \rangle$ is inner product; $\alpha_t(i)$ is the attention paid to source position i at decoder step t .

Step 2 — context vector (weighted sum of encoder states):

$$c_t = \sum_{i=1}^m \alpha_t(i) \bar{h}_i$$

Attention: Context Vector & Output

Step 3 — attentional hidden state:

$$\tilde{h}_t = \tanh(W_c [c_t; h_t])$$

where $[c_t; h_t]$ is the concatenation of source-side context vector c_t and target hidden state h_t .

Output prediction:

$$y_t \sim \text{softmax}(W_o \tilde{h}_t)$$

The softmax behaves like a smooth (differentiable) argmax — if one component of $\tilde{h}_t$ dominates, softmax picks it confidently.

Advantage over plain Encoder–Decoder: no need to compress the entire source into a single vector; the decoder selectively focuses on relevant source positions at each step.

Softmax: Why and How?

$$\text{softmax}(x_1, \dots, x_n) = \left(\frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}} \right)_{i=1, \dots, n}$$

- Outputs a **probability distribution** over n classes (all entries ≥ 0 , sum to 1)
- Behaves like a smooth (differentiable) version of argmax
- If $x_i \gg x_j$, $\text{softmax} \rightarrow$ one-hot at position i
- Used at every decoder step to pick the next syllable

In our model, softmax converts $\tilde{h}_t$ into a probability over the syllable vocabulary — typically a few hundred symbols.

Experiments

Experimental Setup

Dataset:

- 500 parallel (CS, SS) pairs — 2,422 CS words / 7,390 SS syllables
- Six motions & ten objects (Figs. 7 & 8)
- Recorded with Microsoft Kinect v1; one native Japanese speaker
- 400 / 100 train–test split (open test)

SMT setup (Moses [20]):

- GIZA++ alignment [31], symmetrized by grow-diag-final-and [21]
- Lexicalized reordering: msd-bidirectional-fe [44]
- 5-gram LM with interpolated modified Kneser–Ney (SRILM [40, 3])
- MERT tuning [29]; Moses 2.1.1 decoder

RNN setup:

- Chainer v1.15.0.1 [Python]; 1,000 epochs for both neural models

Six Motions & Ten Objects

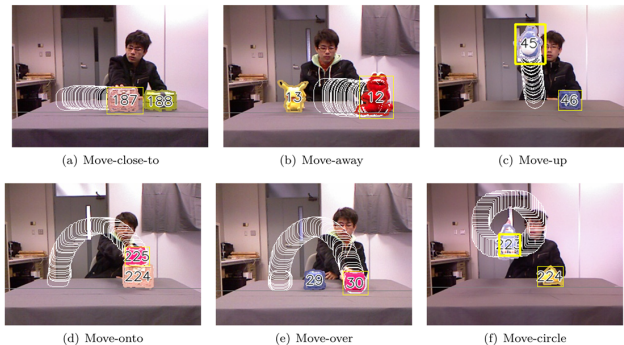


Fig. 7 — Six motions:

close-to, away, up, onto, over, circle

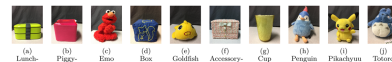


Fig. 8 — Ten objects

BLEU [32] — n-gram precision with brevity penalty:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$

RIBES [14] — rank-based intuitive measure; better for [distant language pairs](#) with very different word order (e.g. CS $\leftrightarrow$ SS).

WER (Word/Syllable Error Rate) via SCLITE (NIST SCTL 2.4.10):

$$\text{WER} = \frac{I + D + S}{N} \times 100$$

- I, D, S — insertions, deletions, substitutions
- N — total reference words/syllables
- Levenshtein alignment between hypothesis and reference

Results

BLEU Scores (Table 2)

Method	CS $\rightarrow$ SS			SS $\rightarrow$ CS		
	Baseline	+ASR	+Object	Baseline	+ASR	+Object
PBSMT	79.68	71.36	77.61	46.69	46.91	44.88
HPBSMT	79.68	70.71	77.83	46.69	43.50	45.10
Enc-Dec	<u>100.00</u>	<u>81.93</u>	<u>99.22</u>	<u>100.00</u>	<u>99.28</u>	<u>100.00</u>
Enc-Dec + Attn	92.53	77.41	88.14	100.00	95.98	100.00

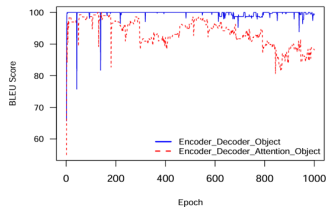
- [Encoder-Decoder](#) wins in every column
- SS $\rightarrow$ CS with object recognition [matches the baseline](#) for both neural models
- ASR errors (88.6% syllable acc.) hurt CS $\rightarrow$ SS noticeably; SS $\rightarrow$ CS is robust

RIBES Scores (Table 3)

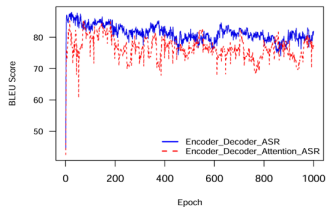
Method	CS $\rightarrow$ SS			SS $\rightarrow$ CS		
	Baseline	+ASR	+Object	Baseline	+ASR	+Object
PBSMT	0.9636	0.9526	0.9621	0.9169	0.9182	0.9130
HPBSMT	0.9636	0.9477	0.9618	0.9169	0.9176	0.9137
Enc-Dec	<u>1.0000</u>	<u>0.9710</u>	<u>0.9986</u>	<u>1.0000</u>	<u>0.9995</u>	<u>1.0000</u>
Enc-Dec + Attn	0.9843	0.9577	0.9778	1.0000	0.9938	1.0000

- Same trend as BLEU — [Encoder-Decoder](#) dominates
- RIBES rewards correct word ordering, especially important here where avg. lengths differ greatly (CS: 4.83 words; SS: 14.54 syllables)

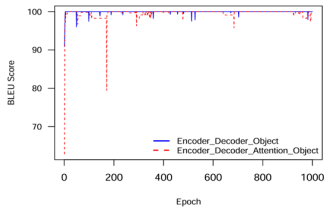
Learning Curves: Enc-Dec vs. Enc-Dec + Attention



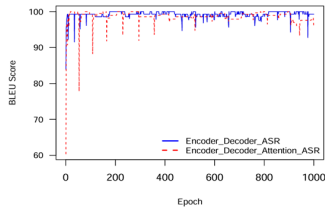
(a) Conceptual structure to Japanese syllable sequence conversion with object recognition results



(b) Conceptual structure to Japanese syllable sequence conversion with ASR results



(c) Japanese syllable sequence to Conceptual structure conversion with object recognition results



(d) Japanese syllable sequence to Conceptual structure conversion with ASR results

Fig. 9 — Encoder-Decoder and Encoder-Decoder with Attention results over 1,000 epochs

Speech Recognition Confusion Pairs

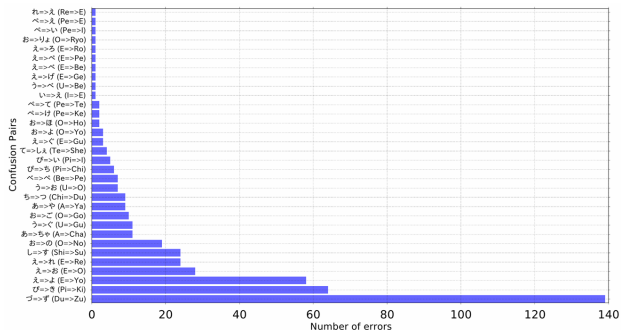


Fig. 10 — Confusion pairs of automatic speech recognition

Top-10 confusion pairs:

1. Du → Zu
2. Pi → Ki
3. E → Yo
4. E → O
5. E → Re
6. Shi → Su
7. O → No
8. A → Cha
9. U → Gu
10. O → Go

Some confusions reflect Japanese phonology (e.g. Du→Zu) — must be reconsidered when extending to other languages.

ASR Errors Can Still Be Learned



Fig. 11 — ASR error on “Move-up Totoro”

Reference: “Totoro o mochiagete”

ASR output: to to ro o mo tsu a ge te



Correct CS: TOTORO TRJ MOCHIAGERU

- Seq-to-seq model can learn from noisy ASR output
- No ASR engine is 100% accurate on real-world speech
- Robustness to ASR error is essential for online acquisition

Discussion & Conclusion

- CS sentences are short (avg. 4.83 words); SS are long (avg. 14.54 syllables) $\Rightarrow$ both BLEU and RIBES are needed
- **SS $\rightarrow$ CS with object recognition** — neural models reach baselines on both BLEU and RIBES
- **CS $\rightarrow$ SS with object recognition** — also comparable to baselines
- Experiments with ASR-output SS are comparable to baseline, but the 88.6% syllable accuracy is a practical concern
- Motion recognition errors **not included** in this study — L-Core is mature (96.8%)
- **Future work:** combine object + motion + speech recognition errors to simulate true online acquisition

- Demonstrated robot language acquisition **without language-specific knowledge**
- End-to-end pipeline: video + utterance $\rightarrow$ CS $\leftrightarrow$ SS
- Encoder–Decoder achieves **baseline performance** on several settings
- Detailed analysis of object and speech recognition errors on 500 videos
- **Future:**
 - Integrate all recognition errors for online acquisition
 - Extend to other languages (English, Chinese, Myanmar)

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Co-authors: Kenta Takabuchi, Kaisei Fukai, Naoto Iwahashi, Takeo Kunishima
Artificial Intelligence Laboratory, Okayama Prefectural University, Japan

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References

-  F. Braune, A. Gojun, A. Fraser. Long-distance reordering during search for hierarchical phrase-based SMT. EAMT 2012, pp. 177–184.
-  A. Cangelosi, M. Schlesinger. *Developmental Robotics: From Babies to Robots*. MIT Press, 2014.
-  S. F. Chen, J. Goodman. An empirical study of smoothing techniques for language modeling. ACL 1996, pp. 310–318.
-  D. Chiang. Hierarchical phrase-based translation. *Comput. Linguist.*, 33(2):201–228, 2007.

-  K. Cho et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. arXiv:1406.1078, 2014.
-  J. Deng et al. ImageNet: A large-scale hierarchical image database. CVPR 2009.
-  T. G. Dietterich, G. Bakiri. Solving multiclass learning problems via error-correcting output codes. *JAIR*, 2:263–286, 1995.
-  J. Donahue et al. DeCAF: A deep convolutional activation feature for generic visual recognition. arXiv:1310.1531, 2013.
-  M. G. Dyer, V. I. Nenov. Learning language via perceptual/motor experiences. CogSci 1993, pp. 400–405.

-  J. Fürnkranz. Round robin classification. *JMLR*, 2:721–747, 2002.
-  A. L. Gorin, S. E. Levinson, A. Sankar. An experiment in spoken language acquisition. *IEEE Trans. Speech & Audio Proc.*, 2(1):224–240, 1994.
-  T. Haoka, N. Iwahashi. Learning of the reference-point-dependent concepts on movement for language acquisition. *IEICE PRMU*, 100(442):39–46, 2000.
-  S. Hochreiter, J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
-  H. Isozaki, T. Hirao, K. Duh, K. Sudoh, H. Tsukada. Automatic evaluation of translation quality for distant language pairs. *EMNLP 2010*, pp. 944–952.

-  N. Iwahashi. Language acquisition through a human–robot interface by combining speech, visual, and behavioral information. *Inf. Sci.*, 156(1–2):109–121, 2003.
-  N. Iwahashi. Robots That Learn Language: Developmental Approach to Human–Machine Conversations. Springer, 2006, pp. 143–167.
-  N. Iwahashi, K. Sugiura, R. Taguchi, T. Nagai, T. Taniguchi. Robots that learn to communicate: A developmental approach. AAAI Fall Symposium 2010.
-  G. James, T. Hastie. The error coding method and PICTS. *JCGS*, 7(3):377–387, 1998.

-  Y. Jia et al. Caffe: Convolutional architecture for fast feature embedding. ACM MM 2014, pp. 675–678.
-  P. Koehn, B. Haddow. Edinburgh's submission to all tracks of the WMT2009 shared task. WMT 2009, pp. 160–164.
-  P. Koehn, F. J. Och, D. Marcu. Statistical phrase-based translation. HLT-NAACL 2003.
-  P. Koehn, F. J. Och, D. Marcu. Statistical phrase-based translation. HLT-NAACL 2003.
-  R. W. Langacker. *Foundations of Cognitive Grammar, Vol. 2*. Stanford Univ. Press, 1991.

-  A. Lee, T. Kawahara, K. Shikano. Julius — an open source real-time large vocabulary recognition engine. EUROSPEECH 2001, pp. 1691–1694.
-  M.-T. Luong, H. Pham, C. D. Manning. Effective approaches to attention-based neural machine translation. EMNLP 2015, pp. 1412–1421.
-  T. Nakamura, T. Nagai, K. Funakoshi, S. Nagasaka, T. Taniguchi, N. Iwahashi. Mutual learning of an object concept and language model based on MLDA and NPYLM. IROS 2014, pp. 600–607.
-  T. Nakamura et al. (same title). IROS 2014, pp. 600–607.
-  N. Iwahashi, R. Taguchi, K. Sugiura, K. Funakoshi, M. Nakano. Robots that learn to converse. NCMMSC2009, pp. 532–537.

-  F. J. Och. Minimum error rate training for statistical machine translation. ACL 2003.
-  F. J. Och, D. Marcu. Statistical phrase-based translation. 2003, pp. 127–133.
-  F. J. Och, H. Ney. Improved statistical alignment models. ACL 2000, pp. 440–447.
-  K. Papineni, S. Roukos, T. Ward, W. Zhu. BLEU: a method for automatic evaluation of machine translation. IBM Research Report RC22176, 2001.
-  D. Roy. Semiotic schemas: A framework for grounding language in action and perception. *Artif. Intell.*, 167(1):170–205, 2005.

-  D. K. Roy, A. P. Pentland. Learning words from sights and sounds: a computational model. *Cognitive Science*, 26(1):113–146, 2002.
-  B. Scholkopf, A. J. Smola. *Learning with Kernels*. MIT Press, 2001.
-  L. Specia. Tutorial: fundamental and new approaches to SMT. RANLP 2011.
-  L. Steels. Evolving grounded communication for robots. *Trends in Cognitive Sciences*, 7(7):308–312, 2003.
-  I. Steinwart, A. Christmann. *Support Vector Machines*. Springer, 2008.
-  K. Stenning. Review of Regier’s “The Human Semantic Potential”. *JLLI*, 10(2):266–269, 2001.

-  A. Stolcke. SRILM — an extensible language modeling toolkit. ICSLP 2002, vol. 2, pp. 901–904.
-  K. Sugiura, N. Iwahashi, H. Kashioka, S. Nakamura. Learning, generation and recognition of motions by reference-point-dependent probabilistic models. *Advanced Robotics*, 25(6–7):825–848, 2011.
-  I. Sutskever, O. Vinyals, Q. V. Le. Sequence to sequence learning with neural networks. arXiv:1409.3215, 2014.
-  T. Taniguchi, T. Nagai, T. Nakamura, N. Iwahashi, T. Ogata, H. Asoh. Symbol emergence in robotics: A survey. arXiv:1509.08973, 2015.

-  C. Tillmann. A unigram orientation model for statistical machine translation. HLT-NAACL Short Papers 2004, pp. 101–104.

Thank you!

Questions & Discussion

Dr. Ye Kyaw Thu

Language Understanding Lab, Myanmar

Visiting Professor, Core-AI Lab, NECTEC, Thailand